A Stationary-Action Principle for Real Quantum Mechanics, and the Equation of Motion of the Free Boundary

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Abstract

Real Quantum Mechanics (RealQM) represents a system of N electrons by N charge densities carried on *non-overlapping* domains of ordinary three-dimensional space, the domains meeting at free boundaries whose position is part of the solution. Its stationary states are obtained by minimising the Coulomb energy over both the densities and the domains. We propose that the *time-dependent* theory is governed by a single *stationary-action* principle in which the dynamical variables are the fields *and* the moving free boundaries, and we carry out the variation. Varying the fields returns the complex time-dependent RealQM (Schrödinger) equations; varying the domains returns the law of motion of the free boundary. Because the action is invariant under time translation and under an independent phase rotation of each domain, Nöther's theorem guarantees, for free, both conservation of the total energy and conservation of the charge *of each electron separately.* The latter fixes the character of the interface: it is a *material* boundary across which no charge flows, so it moves with the normal charge-fluid velocity, with that velocity continuous across it. This is the dynamical expression of RealQM's individuation of electrons by the regions they occupy. The static minimiser is recovered as the stationary-state limit. We give the boundary law explicitly, note that it corrects two naive prescriptions (Dirichlet with slope-matching; Neumann with density-continuity) that violate one conservation law or the other, and delimit where it is needed — real-time, non-adiabatic multi-domain dynamics — and where it is not (Born–Oppenheimer chemistry; the timing of radioactive decay).

1 Introduction

RealQM reformulates quantum mechanics for an N-electron system not as one complex amplitude on a $3N$-dimensional configuration space but as N real-space charge densities [1]. Each electron i is a density ψ_i^2 carried on its own region $\Omega_i \subset \mathbb{R}^3$, the regions non-overlapping, $\Omega_i \cap \Omega_j = \varnothing$; the total charge density is $\rho = \sum_i q_i \psi_i^2$ with $q_i = -1$, and the ground state minimises the Coulomb energy over the fields and the domains, the free boundaries $\partial\Omega_i$ being determined by the minimisation. This static variational formulation reproduces atomic and molecular structure, and — read for protons and electrons at femtometre scale — nuclear binding, all from electrostatics [2].

Two things are then missing. First, a *dynamical* principle: what equation governs the fields, and — the question we settle here — what governs the *motion of the free boundary* in time? Second, the assurance that the dynamics respects the conservation laws it must: the energy, the total charge, and, since the electrons are distinct objects individuated by their domains, the charge of *each* electron. We show that a single stationary-action principle supplies all of this at once. The free boundary is not an auxiliary construction with an *ad hoc* rule of motion; it is a dynamical variable of the same action that gives the field equations, and its law of motion follows by variation, with the conservation laws guaranteed by the symmetries of the action.

2 The action

Work in atomic units ($\hbar = 1$, $m_e = 1$). The dynamical variables are the complex fields $\psi_i(x,t)$, each supported on a time-dependent domain $\Omega_i(t)$, and the domains themselves (equivalently their boundaries). Take the action

$$S = \int dt \left\{ \sum_i \int_{\Omega_i(t)} \Big[\tfrac{i}{2}\big(\psi_i^* \partial_t \psi_i - \partial_t \psi_i^* \, \psi_i\big) - \tfrac{1}{2} |\nabla \psi_i|^2 \Big] dx - \tfrac{1}{2} \iint \frac{\rho(x)\rho(y)}{|x-y|} \, dx \, dy \right\}, \tag{1}$$

with $\rho = \sum_i q_i |\psi_i|^2$ and the constraints $\Omega_i \cap \Omega_j = \varnothing$ and, for each i, $\int_{\Omega_i} |\psi_i|^2 = 1$ (unit charge per electron). The first bracket is the standard Schrödinger Lagrangian density for each domain; the last term is the RealQM Coulomb energy, written as a potential energy in the action. We take (1) to be the basic principle of the theory: the physical evolution is a stationary point of S under variation of the fields and of the domains.

3 Field variation: the RealQM equations

Vary ψ_i^* at fixed domains. The bulk Euler–Lagrange equation is the complex time-dependent RealQM (Schrödinger) equation

$$i\,\partial_t \psi_i = -\tfrac{1}{2} \nabla^2 \psi_i + q_i \, \phi[\rho] \, \psi_i, \qquad \phi[\rho](x) = \int \frac{\rho(y)}{|x-y|} \, dy, \tag{2}$$

ϕ being the self-consistent Coulomb potential of the total density. The variation also leaves a term on the free boundary, $-\int_{\partial\Omega_i} \mathrm{Re}(\partial_n \psi_i^* \, \delta\psi_i)$; its treatment is dictated by the domain variation of the next section, to which it is naturally paired.

4 Domain variation: the free boundary is a material interface

Now vary the domains. It is enough to consider two domains sharing an interface $\Gamma(t)$ with unit normal n pointing from Ω_1 into Ω_2; let the interface move with normal speed V_n. Two facts about (1), both consequences of its symmetries, fix the interface law completely.

Per-electron charge conservation. The action (1) is invariant under an *independent* global phase rotation of each field, $\psi_i \to e^{i\alpha_i} \psi_i$ (the Coulomb term depends only on the densities $|\psi_i|^2$). By Nöther's theorem each domain carries a conserved charge, and the accompanying local law is the continuity equation

$$\partial_t |\psi_i|^2 + \nabla \cdot \mathbf{j}_i = 0, \qquad \mathbf{j}_i = \mathrm{Im}(\psi_i^* \nabla \psi_i). \tag{3}$$

The charge of electron i is $Q_i = \int_{\Omega_i(t)} |\psi_i|^2$, and with the boundary moving,

$$\frac{dQ_i}{dt} = \int_{\Omega_i} \partial_t |\psi_i|^2 \, dx + \int_{\partial\Omega_i} |\psi_i|^2 \, V_n \, dS = \int_\Gamma \big(V_n \, |\psi_i|^2 - \mathbf{j}_i \cdot n \big) \, dS, \tag{4}$$

using (3) and vanishing current on the outer boundary. For Q_i to be conserved — for each electron to keep its unit charge, as its very identity in RealQM requires — the integrand must vanish pointwise on Γ:

$$\boxed{V_n = \frac{\mathbf{j}_i \cdot n}{|\psi_i|^2} = \mathbf{v}_i \cdot n} \tag{5}$$

where $\mathbf{v}_i = \mathbf{j}_i / |\psi_i|^2 = \nabla S_i$ is the charge-fluid (Madelung) velocity of domain i, with $\psi_i = |\psi_i| e^{iS_i}$. Equation (5) says the interface is a *material* boundary: no charge crosses it, and it is carried along by the flow. Applying it to *both* domains forces the velocities to agree on the interface,

$$\mathbf{v}_1 \cdot n = \mathbf{v}_2 \cdot n = V_n \quad \text{on } \Gamma, \tag{6}$$

so that a single boundary speed is well defined. Equations (5)–(6) are the law of motion of the free boundary: it advects with the (common) normal charge-fluid velocity.

The paired field condition. The boundary term left by the field variation (§3) is now fixed consistently: on a material interface the natural condition is that the fields carry no current *through* the boundary in its own frame, which (5) already enforces; the residual variation vanishes with the density and its normal flux matched by (6). No charge is created, destroyed, or exchanged at Γ; the two clouds remain in contact and simply move together.

5 Conservation of energy and momentum

Because (1) carries no explicit time dependence, it is invariant under time translation, and Nöther's theorem gives a conserved energy

$$E = \sum_i \int_{\Omega_i(t)} \tfrac{1}{2} |\nabla \psi_i|^2 \, dx + \tfrac{1}{2} \iint \frac{\rho\rho}{|x-y|}, \tag{7}$$

exactly, for the coupled field–boundary evolution — provided the boundary obeys its own Euler–Lagrange law (5) rather than an externally imposed rule. This is the crux: energy conservation is not an extra property to be checked but a structural consequence of deriving the boundary motion from the same action as the fields. Likewise, invariance under spatial translation gives conserved total momentum $\mathbf{P} = \sum_i \int_{\Omega_i} \mathrm{Im}(\psi_i^* \nabla \psi_i)$. Thus one principle delivers, together, the field equations, the boundary law, and the conservation of energy, momentum, and per-electron charge.

6 The static limit, and two naive prescriptions it corrects

For a stationary state $\psi_i(x,t) = \varphi_i(x) e^{-i\varepsilon_i t}$ the currents $\mathbf{j}_i$ vanish, so by (5) $V_n = 0$: the free boundary is at rest, and (1) reduces to the static Coulomb-energy minimisation over fields and domains that defines the RealQM ground state. The action principle is therefore the dynamical parent of the static formulation used for atoms, molecules and nuclei [1, 2], not a separate construction.

The material-interface law also settles, and corrects, two prescriptions one is tempted to impose by hand. A *Dirichlet* interface ($\psi_i = 0$ on Γ) with the boundary placed where the normal slopes match, $|\partial_n\psi_1| = |\partial_n\psi_2|$, conserves per-electron charge but is *not* the stationary-action law and does not, in general, conserve energy. A *Neumann* interface ($\partial_n\psi_i = 0$) placed where the densities are continuous, $|\psi_1| = |\psi_2|$, can be arranged to conserve energy but transfers charge across the moving boundary and so *violates* per-electron charge conservation. Only the material law (5)–(6), in which the boundary moves with the flow, respects both conservation laws, precisely because both descend from the one action.

7 Relation to the Bernoulli condition of static RealQM

Static RealQM equips the free boundary with a *Bernoulli condition*: continuity of the fields across the interface, $\psi_1 = \psi_2$ on Γ (forced by $\Psi = \sum_i \psi_i \in H^1$), together with the homogeneous Neumann condition $\partial_n\psi_i = 0$, the latter arising as the transversality condition when the Coulomb energy is minimised over the domains [1]. The two formulations meet, and it is worth being exact about how.

In the *static* limit they coincide. For a stationary state the currents $\mathbf{j}_i$ vanish, so (5) gives $V_n = 0$: the boundary is at rest, and (1) reduces to the very energy minimisation whose transversality condition is the Neumann law. The stationary-action principle therefore *recovers* the Bernoulli condition of static RealQM in exactly the regime where static RealQM is used — atoms, molecules, and nuclei.

Out of equilibrium the two must be distinguished, and there are two genuine subtleties.

First, the *density-continuity* half of the Bernoulli condition, $\rho_1 = \rho_2$ on Γ, does *not* appear among the equations derived above. It is not a consequence of the domain variation but a constraint on the admissible function space, imposed in static RealQM by requiring $\Psi = \sum_i \psi_i \in H^1$. What the variation yields at the interface is instead *velocity* continuity, (6), from per-electron charge conservation. These are different conditions: a material interface permits a density jump (a contact discontinuity), which would violate $\Psi \in H^1$. Requiring both — the H^1 density continuity of static RealQM and per-electron charge conservation — with the boundary in motion is not automatic, and reconciling them is precisely the open point.

Second, and relatedly, the static Neumann condition applied literally in real time would force $\mathbf{j}_i \cdot \mathbf{n} = \mathrm{Im}(\psi_i^* \partial_n \psi_i) = 0$ and hence, by (5), $V_n = 0$: a *moving* material boundary is incompatible with strict Neumann. Neumann is a property of the *static* transversality; the time-dependent transversality condition — from including the temporal terms of (1) in the domain variation — differs from it and permits motion. Per-electron charge conservation fixes the *kinematics* of the moving boundary (it is material, (5)); the precise dynamic field condition that replaces static Neumann and density continuity for a genuinely moving interface is a point we identify but do not settle here. In the regime where RealQM has so far been applied — ground states and Born–Oppenheimer relaxation — neither distinction arises, since there the boundary is at rest, the currents vanish, and the Bernoulli condition (density continuity plus Neumann) holds.

These two subtleties are in fact one choice, and it is worth stating it sharply. The material interface (5)–(6) is, in the language of fluid mechanics, a *contact discontinuity*: the normal velocity is continuous across it while the density may jump. Time-dependent RealQM must then adopt one of two positions. It is worth being clear that density continuity is nothing exotic: since $\rho_i = |\psi_i|^2$, continuity of the density is simply continuity of ψ^2, which follows at once from ψ being continuous, i.e. from $\Psi \in H^1$. Continuity of ψ fixes its *value* at the interface, and hence $\rho_1 = \rho_2$; the velocity, by contrast, is the phase *gradient* $\mathbf{v} = \nabla S$, and continuity of ψ says nothing about the gradient. Density continuity (of the value) and velocity continuity (of the gradient) are therefore independent conditions. The two positions are then:

(A) *Retain* $\Psi \in H^1$. This is the natural, economical condition — the wavefunction is continuous, so $\rho_1 = \rho_2$ holds automatically. The question is whether one may *also* impose per-electron charge conservation, i.e. the velocity continuity (6): that is a condition on the gradient, over and above the continuity of the value, and requiring both may *over-determine* the interface, since density continuity does not by itself persist under the material law (5).

(B) *Give up* ψ*-continuity to conserve charge.* The interface becomes a genuine contact discontinuity: per-electron charge conservation and (6) govern it, a jump in ψ (hence in ρ) is permitted, and Ψ is no longer strictly H^1. This is the charge-conserving choice; it reduces to the static Bernoulli condition at rest, where $\mathbf{v}_i = 0$ makes velocity continuity trivial and density continuity is the substantive statement.

So the fork is between keeping the wavefunction continuous (A), at the risk that per-electron charge conservation cannot then also be imposed, and enforcing charge conservation (B), at the price of a jump in the wavefunction. We record it and leave its resolution — a genuine choice about the time-dependent theory — open.

8 Scope

The boundary law matters exactly where the domains genuinely rearrange *in time*. In *Born–Oppenheimer* chemistry the electrons re-minimise at each nuclear geometry, using the static free boundary of §6; the time-dependent law is not invoked, and ordinary reaction paths, barriers and binding energies are unaffected. It is needed for *real-time, non-adiabatic* multi-domain dynamics — attosecond electron motion, charge migration, photochemistry, collisions — where how the electron territories reshape in time is itself the physics. It is, by contrast, immaterial to phenomena that are insensitive to the partition: the timing and rate of radioactive decay, for instance, are carried by the total charge density tunnelling a barrier and do not depend on the interior boundary between domains at all.

9 Conclusion

RealQM is governed by a single stationary-action principle, (1), whose dynamical variables are the charge-density fields and the free boundaries between them. Varying the fields gives the complex time-dependent RealQM equations; varying the domains gives the law of motion of the free boundary, which the independent phase symmetry of each domain fixes to be a *material* interface, advected by the common normal charge-fluid velocity, (5)–(6). Energy, momentum and per-electron charge are then conserved automatically, by Nöther's theorem, because all of it descends from one symmetric action. The static energy minimisation that defines atoms, molecules and nuclei is the stationary-state limit of the same principle. The free boundary, in short, is not an appendage to RealQM but one of its dynamical variables, and it moves — like everything else in the theory — by stationary action.

References

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